

Estimating Rental Yields from Unpaired Online Housing Listings

A Hierarchical Bayesian Approach with Evidence from Armenia

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August 29, 2026

Abstract

Housing-market monitoring is difficult in settings where official transaction and rental microdata are limited, while online classifieds provide abundant but compositionally mismatched observations. We develop a multivariate hierarchical Bayesian framework that estimates composition-adjusted rental yields from unpaired online listings, where rent and sale records refer to different dwellings. Rent and sale prices are modeled as separate log-linear hedonic surfaces linked through location intercepts drawn from a bivariate normal whose rent–sale correlation ρ is learned from the data. Dwelling characteristics absorb composition differences that bias raw median rent-to-price ratios, and yields are recovered as posterior distributions for a standardized apartment profile. Using 8,863 listings across 36 Armenian districts, the fitted model shows good in-sample posterior predictive calibration: 91.3% of rent observations and 90.9% of sale observations fall within their nominal 90% posterior predictive intervals, with no NUTS divergent transitions. Estimated yields range from 5.4% in Kentron to 8.7% in Jermuk and differ materially from naive median ratios; sale prices scale almost twice as fast as rents with size ($\beta_{\text{sqm}}^P \approx 0.90$ vs. $\beta_{\text{sqm}}^R \approx 0.52$), while observed condition differences are capitalized more strongly in asking rents than in asking sale prices. The framework requires neither matched-pair nor registry data and offers a transferable way to extract housing-market signals from online microdata in data-constrained markets.

1 Introduction

Housing-market surveillance depends on reliable measures of both asset prices and the income those assets can generate. Yet in many markets, especially outside the largest and best-documented cities, detailed transaction and rental microdata are limited or unavailable. Online classifieds offer a potentially rich alternative source of housing microdata, but extracting policy-relevant signals from them requires confronting how the observed rental and sale populations differ. Gross rental yield, closely related to the inverse price-to-rent ratio, is a useful example: it summarizes rental income relative to the capital required to purchase housing and can support comparisons across locations, housing segments, and time when measured consistently.

The standard gross yield reported by property-data aggregators divides a market’s median monthly rent by its median sale price and multiplies by twelve (Global Property Guide, 2026). This procedure quietly assumes that rental and sale observations represent comparable dwelling populations. In practice, they often do not: the two pools can differ systematically in size, room count, condition, furnishing, and other characteristics (Glaeser and Gyourko, 2007; Hill and Syed, 2016; Bracke, 2015). A raw ratio can therefore move because the underlying housing mix differs, rather than because the rent–price relationship itself differs. The measurement problem is especially relevant when researchers rely on public online listings instead of matched administrative or transaction records.

Existing work does not directly address this data setting. Hedonic housing models overwhelmingly estimate sale prices, while rental hedonic models are less common because rental microdata are less frequently captured in government statistical reporting. Studies that estimate price-to-rent ratios at the property or neighborhood level typically rely on matched rent–sale records, where the same dwelling or building can be observed on both sides of the market. Macro-level studies take the opposite approach, using pre-aggregated city or national indices to study broad price-to-rent dynamics. These approaches are useful, but they do not solve the setting considered here: recovering composition-adjusted rental yields across nested geographic levels from unpaired online rent and sale listings while retaining dwelling-level housing characteristics.

This paper develops a generative framework that fills that gap. Rent and sale listings are modeled as linked but distinct log-linear hedonic surfaces. Each location receives a paired latent intercept $(\alpha_\ell^R, \alpha_\ell^P)$ drawn from a bivariate normal whose correlation ρ is learned, so that locations sparse on one side borrow strength from the other through the correlation rather than through arbitrary extrapolation. Per-listing covariates (size, rooms, renovation level, furniture status) enter on both sides with separate coefficients, absorbing the composition gap that biases naive medians. Yields are derived draw-by-draw from the posterior for a standardized apartment profile, returning a distribution rather than a point estimate.

This paper makes three connected contributions.

1. **A composition-adjusted yield estimand for unpaired online listing markets.** Rather than treating rental yield as a ratio of aggregate medians, we define it as the pos-

terior distribution of the yield for a single standardized dwelling profile, a quantity with posterior uncertainty rather than a point estimate. The estimand builds on the hedonic price–rent literature’s emphasis on quality adjustment but adapts it to a setting where rent and sale listings are unpaired, drawn from different dwelling pools, and observed through public online classifieds rather than matched transaction or registry data.

2. **A multivariate hierarchical Bayesian model for linked rent and sale surfaces.** We estimate separate hedonic surfaces for rents and sale prices and couple their location effects through a bivariate-normal hierarchy whose rent–sale correlation is learned from the data. This learned correlation, rather than any physical matching of dwellings, is what lets sparse locations borrow strength across both sides of the market, and the model returns a full posterior distribution for every implied yield.
3. **An empirical application to housing-market measurement in Armenia.** We apply the framework to apartment rental and sale listings from list.am, producing district-level implied yield estimates in a market where official housing microdata are limited. The application shows how public classifieds can be converted into a composition-adjusted housing-market indicator, and demonstrates that naive median ratios can materially misstate local yields when rent and sale listing pools differ.

2 Related Literature

The paper sits at the intersection of four largely separate literatures.

Hedonic price modeling. The hedonic tradition, formalized by Rosen (1974) and surveyed comprehensively by Sirmans et al. (2005) and Hill (2013), models log price as a function of dwelling and location characteristics. The literature is vast and methodologically mature on the sale side, with extensive treatments of functional form, spatial heterogeneity, and time-varying coefficients. We borrow the log-linear hedonic structure directly, but for an objective that hedonic models rarely target: implied yield rather than price.

Rental hedonic models. A smaller body of work applies hedonic methods to rents. Chapelle and Eyméoud (2022) emphasize a central data asymmetry: while housing prices are often systematically recorded, accurate rent datasets are much less widely available. They address this gap by collecting online rental advertisements in France, validating the resulting data against conventional rent sources, and estimating hedonic rent and price models to document spatial variation in rent-price ratios. Rental hedonics in developing-economy and post-Soviet contexts are rarer still; Stepanyan et al. (2010) estimates a panel of macro determinants of house prices in former-Soviet countries but does not separately model rents.

The rent side is therefore the harder measurement problem in the setting considered here, which is precisely where online listings have a comparative advantage.

Joint rent–price and yield estimation. The literature most directly relevant to ours is also the thinnest. Bracke (2015) estimates price-to-rent ratios from a matched dataset of central London apartments observed on both sides of the market within a short window. Hill and Syed (2016) extend the hedonic framework to construct hedonic price–rent ratios for Sydney, again leveraging linkable transaction–rental records. Chen and Ni (2011) estimate property-specific price-to-rent ratios in Shanghai and Shenzhen using a Bayesian approach, again on matched units. The overarching theme in this cross-section of literature is matched-pair data: the same dwelling (or building) observed for both rent and sale. Such data are rare outside the largest and best-documented urban markets, and the methodology does not extend naturally to settings where rent listings and sale listings refer to fundamentally different dwellings. Lee and Park (2018) estimate rent-to-price ratios at the micro-spatial scale using aggregated zonal indices, which is a different design that loses property-level adjustment. We depart from both strategies: instead of linking the two sides of the market through matched records or collapsing them into zonal aggregates, we couple dwelling-level rent and sale hedonic surfaces through a learned latent correlation between their location effects, recovering the rent–sale relationship without any paired, linkable, or address-matched observations.

Country-scale yield and online-listing work. A separate body of macroeconomic work studies yield dynamics at the country or city level using pre-aggregated public indices (e.g. official transaction registers, OECD or BIS housing series) regressed on macroeconomic drivers such as interest rates, demographics, and credit. These studies are informative for cross-country comparison but operate on data already aggregated above the location of interest here. Online listings have, separately, been used as a real-time input to housing-market prices and indices: Lyons (2019) examines whether asking prices proxy transaction prices in distressed markets, and Loberto et al. (2022) build a high-frequency big-housing-data index for Italy from listings, focusing on prices rather than the gap between prices and rents. Rae (2015) surveys other online-listing applications. While these studies confirm the viability of using online listing data, we are not aware of work that constructs sub-national yield posteriors from unpaired rent and sale listings via a generative model.

Bayesian hierarchical modeling. The multilevel statistical modelling we use is well-established (Gelman et al., 2013; McElreath, 2020), including the bivariate-normal random-effects structure, the LKJ prior on correlation matrices (Lewandowski et al., 2009), the non-centered Cholesky parameterization that stabilizes sampling in the presence of high correlation, and the No-U-Turn Sampler implementation (Hoffman and Gelman, 2014). We rely on the PyMC implementa-

tion (Salvatier et al., 2016). None of these components are novel; the combination, applied to this measurement problem, is.

Differentiation. Against this literature, our framework occupies an unfilled cell: it estimates composition-adjusted, location-level yields with posterior uncertainty from unpaired online listings, replacing the physical dwelling-matching that prior micro-level studies depend on with a learned latent correlation between rent and sale surfaces.

3 Data

The empirical demonstration uses apartment listings scraped from the dominant Armenian classifieds platform. The scrape covers two categories: apartments for sale and long-term rentals, across all eleven administrative regions of Armenia. The final dataset is a cross-sectional snapshot collected over a single week in May 2026.

The model uses a deliberately small, strict feature set: listing type, district, advertised price in U.S. dollars, apartment area in square meters, room count, renovation level, and furniture status. Renovation and furniture are extracted from the detail page’s attribute table and have 100% coverage. The many additional fields available from the listing pages, including coordinates, floor, total floors, construction type, balcony, free-text description, and view counts, are excluded from the main specification. The goal is to retain characteristics that are widely reported, interpretable, and available on both the rental and sale sides of the market, making the framework more transferable across countries and listing platforms.

Three cleaning steps are applied. First, listings are dropped if any model field is missing. Second, values are bounded to plausible ranges (rent \$100–\$6,000/mo, sale \$10k–\$2M, area 20–300 m², rooms 1–5), which removes a small fraction of typos and likely mislabeled posts. Third, locations are retained only if they contain at least five rental and five sale listings, so each reported yield is anchored in real data on both sides. Table 1 shows the attrition. The final analytic sample is 8,863 listings spanning 36 districts.

Table 1: Listing counts after cleaning.

Stage	Rent	Sale	Total
Raw listings	3,959	5,558	9,517
After value cleaning	3,920	5,458	9,378
After location filter	3,844	5,019	8,863

Composition between sides differs sharply (Table 2). Rentals are overwhelmingly furnished and concentrate in mid-to-high renovation tiers; sales contain a broader distribution including a substantial share of low-renovation units. This is the empirical basis for not using the raw median ratio as the target estimand.

Table 2: Rent–sale composition differences (share of side, %).

Feature	Rent	Sale
Furnished (“With”)	96.1	44.4
Designer renovation	32.5	10.5
Major renovation	33.1	32.1
No renovation	0.2	15.9
Old renovation	1.5	9.3

4 Methodology

This section defines the composition-adjusted yield estimand and the hierarchical model used to estimate it. The model aims to estimate separate hedonic surfaces for rents and sale prices, link them through location-level effects, and derive rental yield as a posterior quantity for a standardized dwelling profile.

4.1 Sample Construction

The model is estimated on the cleaned analytic sample described in Section 3. The key methodological requirement is that rent and sale listings enter the model with a common set of dwelling characteristics. Accordingly, the same feature definitions, value bounds, location filters, and categorical coding are applied to both sides of the market before estimation.

Renovation and furniture are represented as categorical predictors with Major renovation and With furniture as the reference levels. Apartment area is modeled as $\log(A/60)$ and room count as $(r - 2)$, so that location intercepts correspond to the expected log rent or log sale price of the baseline dwelling profile: a 60 m², 2-room apartment with Major renovation and With furniture. This baseline profile is also used for the main posterior yield estimates reported below.

4.2 Generative model

For rental listing i and sale listing j ,

$$\log R_i \sim \mathcal{N}(\mu_i^R, \sigma_R), \quad (1)$$

$$\log P_j \sim \mathcal{N}(\mu_j^P, \sigma_P), \quad (2)$$

with linear predictors

$$\begin{aligned} \mu_i^R = & \alpha_{\ell[i]}^R + \beta_{\text{sqm}}^R \log \frac{A_i}{60} + \beta_{\text{rm}}^R (r_i - 2) \\ & + \mathbf{x}_{i,\text{reno}}^\top \boldsymbol{\beta}_{\text{reno}}^R + \mathbf{x}_{i,\text{furn}}^\top \boldsymbol{\beta}_{\text{furn}}^R, \end{aligned} \quad (3)$$

$$\begin{aligned} \mu_j^P = & \alpha_{\ell[j]}^P + \beta_{\text{sqm}}^P \log \frac{A_j}{60} + \beta_{\text{rm}}^P (r_j - 2) \\ & + \mathbf{x}_{j,\text{reno}}^\top \boldsymbol{\beta}_{\text{reno}}^P + \mathbf{x}_{j,\text{furn}}^\top \boldsymbol{\beta}_{\text{furn}}^P. \end{aligned} \quad (4)$$

The centering of $\log(A/60)$ and $(r - 2)$ makes the intercept directly interpretable as the expected log rent (resp. log price) of a 60 m², 2-room apartment at the reference renovation/furniture level. The dummy vectors $\mathbf{x}_{\cdot,\text{reno}}$ and $\mathbf{x}_{\cdot,\text{furn}}$

drop the reference category, so the reference combination contributes zero.

The two sides are linked through paired location intercepts:

$$\begin{pmatrix} \alpha_{\ell}^R \\ \alpha_{\ell}^P \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} \mu_{\alpha,R} \\ \mu_{\alpha,P} \end{pmatrix}, \Sigma\right). \quad (5)$$

Let Ω denote the 2×2 correlation matrix for the location effects, with $\rho = \Omega_{RP}$, and let $D_{\alpha} = \text{diag}(\sigma_{\alpha,R}, \sigma_{\alpha,P})$. The covariance matrix is

$$\Sigma = D_{\alpha} \Omega D_{\alpha} = \begin{pmatrix} \sigma_{\alpha,R}^2 & \rho \sigma_{\alpha,R} \sigma_{\alpha,P} \\ \rho \sigma_{\alpha,R} \sigma_{\alpha,P} & \sigma_{\alpha,P}^2 \end{pmatrix}. \quad (6)$$

Σ is parameterized through its Cholesky factor with an LKJ prior on Ω and with the standard non-centered reparameterization (Lewandowski et al., 2009; McElreath, 2020):

$$\Sigma = LL^{\top}, \quad \begin{pmatrix} \alpha_{\ell}^R \\ \alpha_{\ell}^P \end{pmatrix} = \begin{pmatrix} \mu_{\alpha,R} \\ \mu_{\alpha,P} \end{pmatrix} + Lz_{\ell}, \quad z_{\ell} \sim \mathcal{N}_2(0, I).$$

ρ is the parameter of substantive interest at this level: it captures whether locations that are expensive to rent are also expensive to buy.

Prior choices reflect the real USD scale of the data and follow weakly informative Bayesian choices:

$$\begin{aligned} \mu_{\alpha,R} &\sim \mathcal{N}(\log 400, 1), \\ \mu_{\alpha,P} &\sim \mathcal{N}(\log 100,000, 1), \\ \sigma_{\alpha,R}, \sigma_{\alpha,P} &\sim \text{Exp}(1), \\ \Omega &\sim \text{LKJ}(\eta = 2), \quad \rho = \Omega_{RP}, \\ \beta_{\text{sqm}}^R &\sim \mathcal{N}(0.7, 0.4), \\ \beta_{\text{sqm}}^P &\sim \mathcal{N}(0.9, 0.4), \\ \beta_{\text{rm}}^R, \beta_{\text{rm}}^P &\sim \mathcal{N}(0, 0.3), \\ \sigma_R, \sigma_P &\sim \text{Exp}(1), \\ \beta_{\text{reno}}^{R/P}, \beta_{\text{furn}}^{R/P} &\sim \mathcal{N}(0, 0.5). \end{aligned}$$

4.3 Posterior yield estimand

For a standardized apartment profile p at location ℓ , the implied gross annual yield is constructed for every posterior draw s :

$$Y_{\ell,p}^{(s)} = \frac{12 \cdot \exp(\hat{\mu}_{\ell,p}^{R(s)})}{\exp(\hat{\mu}_{\ell,p}^{P(s)})}. \quad (7)$$

The estimand is the posterior distribution of $Y_{\ell,p}$ at the reference profile p^* : 60 m², 2 rooms, Major renovation, With furniture. Alternate profiles vary one input at a time to read off marginal effects on yield.

4.4 Implementation

The model is implemented in PyMC 5.28 (Salvatier et al., 2016) and sampled with NUTS (Hoffman and Gelman, 2014): 4 chains, 2,000 tune + 1,000 draws each, `target_accept` = 0.95, fixed random seed. Posterior predictive draws are generated in-sample for both likelihoods.

5 Results

5.1 Sampler diagnostics

All four chains report zero divergences. The maximum Gelman–Rubin $\hat{R}$ across all monitored parameters is 1.013 for μ_{α} and below 1.01 everywhere else. Bulk effective sample sizes are healthy for the regression slopes and residual scales ($\text{ESS} > 2,500$). The top-level hyperparameters μ_{α} and the two σ_{α} values exhibit lower ESS (~ 250 and ~ 575 respectively), a consequence of the posterior for ρ being concentrated close to the upper boundary, which makes the bivariate covariance geometry harder to explore.

5.2 Posterior predictive checks

In-sample posterior predictive distributions overlay the observed log-rent and log-price densities well (Figure 1). The observed series exhibit small jagged peaks not reproduced by the smooth Gaussian predictive likely due to round-number asking-price clustering by sellers (e.g. 250,000 AMD, 300,000 AMD). The key calibration check is empirical coverage: the share of observed listings that fall inside their model-implied 90% posterior predictive intervals. Coverage is 91.3% for rents and 90.9% for sale prices, close to the nominal 90% target. This indicates that the predictive intervals are close to nominal in-sample, with only slight over-coverage. These in-sample posterior predictive checks do not indicate an immediate need for heavier-tailed alternatives.

Per-location PPC checks (Figure 2) show every district’s predicted mean log rent and log price hugging the observed means closely, with 90% bars crossing the $y = x$ line in all 36 cases. The widest bars correspond exactly to the smallest-sample locations, as expected under partial pooling.

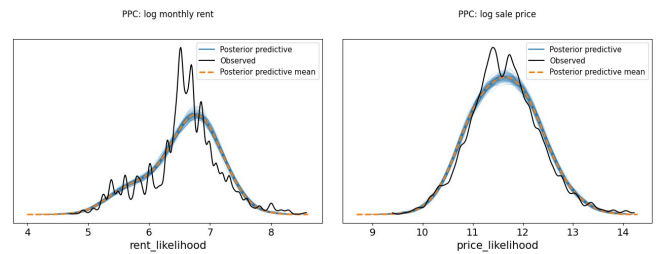


Figure 1: Posterior predictive density overlays for log monthly rent (left) and log sale price (right). Black = observed, blue = posterior predictive draws.

5.3 The learned rent–sale correlation

The posterior median of ρ is 0.987 (95% credible interval [0.96, 1.00]; Figure 3). Across Armenian districts, the rent and sale-price location premia move essentially in lockstep. A single “location quality” factor explains the overwhelming majority of cross-district price variation. This does not imply that yield is constant across locations; it implies that the yield

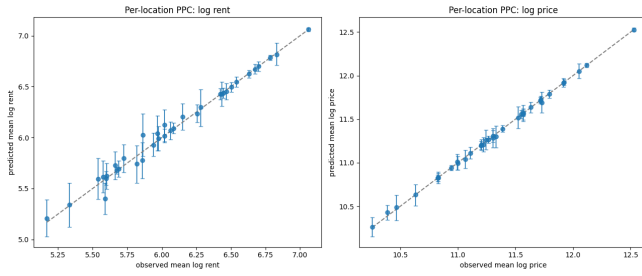


Figure 2: Per-location PPC: observed vs. predicted mean (log scale) for rent (left) and sale (right). Dashed = $y = x$. Bars: 90% interval.

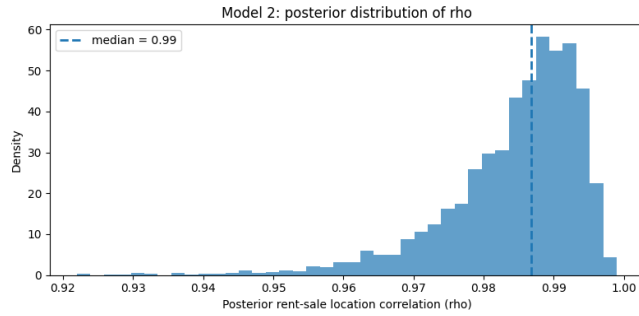


Figure 3: Posterior distribution of the rent-sale location correlation ρ .

signal lives in the small deviations from collinearity rather than in independent geographic variation between the two surfaces. The hierarchy with paired intercepts is what isolates those deviations.

5.4 Size and room elasticities

Size enters log-rent with elasticity 0.521 ($\hat{R} = 1.000$, 95% CI [0.485, 0.560]) and log-price with elasticity 0.904 ([0.875, 0.935]). Doubling apartment area therefore raises rents by a factor of $2^{0.521} \approx 1.43$ but prices by $2^{0.904} \approx 1.87$, revealing a striking asymmetry. The yield ratio falls by approximately 23% on doubling. Room count on the other hand, conditional on area, contributes very little to yield changes ($\beta_{rm}^R = 0.019$, $\beta_{rm}^P = -0.035$). Table 3 reports median yield across locations for four canonical size profiles.

Table 3: Median posterior yield (%) across locations, by apartment size profile.

Profile	Median yield
40 m ² , 1 room	7.40
60 m ² , 2 rooms (baseline)	6.69
90 m ² , 3 rooms	6.04
120 m ² , 4 rooms	5.72

5.5 Condition is rewarded more in rent than in resale

The renovation and furniture coefficient blocks (Figure 4; full table in Appendix A) reveal a substantive asymmetry. A no-renovation listing rents for roughly 42% less than an otherwise comparable Major-renovated unit ($\beta^R = -0.542$), but sells for only 12% less ($\beta^P = -0.133$). Combining the rent and sale effects, the implied yield ratio is $\exp(-0.542 - (-0.133)) \approx 0.66$. More simply, a no-renovation unit is estimated to yield only about 66% as much as an otherwise comparable Major-renovated unit, or roughly one-third less. The pattern repeats across the condition ladder: Old, Partial, and Cosmetic categories penalize rent more than they penalize sale price. Furniture is similar: unfurnished units rent for $\sim 12\%$ less but sell for $\sim 5\%$ less than furnished ones, producing a $\sim 7\%$ yield discount.

Substantively, the rental market prices condition more aggressively than the resale market does. This is consistent with rentals being immediate consumption (tenants pay for what they get now) while resale embeds expectations about future maintenance, taste, and liquidity. The estimates suggest that observed condition differences are capitalized more strongly in asking rents than in asking sale prices.

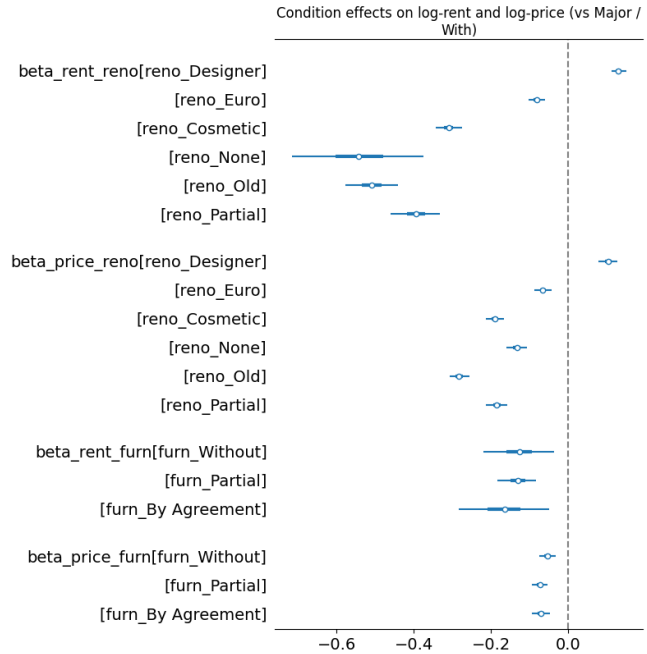


Figure 4: Renovation and furniture effects on log rent (top blocks) and log sale price (bottom blocks), each relative to the reference (Major/With). Means and 94% intervals.

5.6 Implied yields by location

Figure 5 shows posterior median yields and 80% intervals for the baseline profile across all 36 districts. Several patterns emerge. Central Yerevan districts sit at the bottom: Kentron's

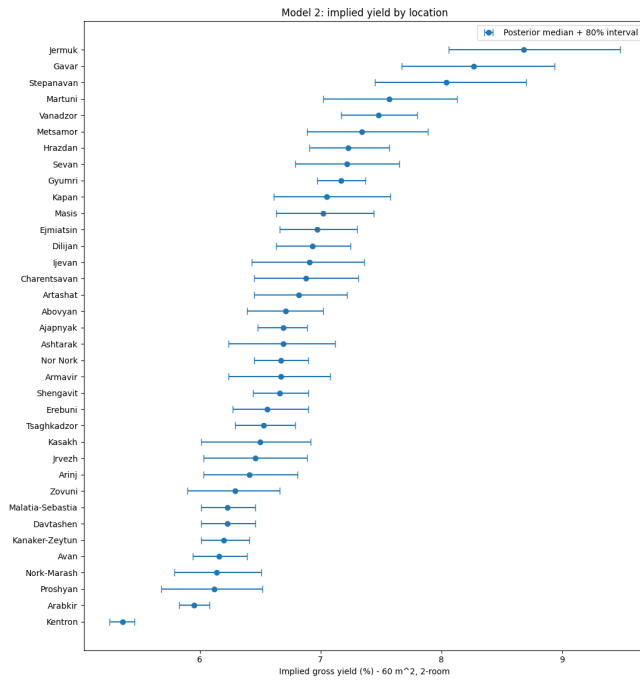


Figure 5: Posterior median + 80% interval of implied gross rental yield by district (60 m², 2-room, Major renovation, With furniture).

median yield is 5.36% with a very tight interval [5.25, 5.46] owing to its large sample; Arabkir follows at 5.95%. Outer-Yerevan and suburban districts (Davtashen, Malatia-Sebastia, Avan, Kanaker-Zeytun) cluster at 6.1–6.3%. Secondary cities and resort towns occupy the middle: Vanadzor 7.48%, Gyumri 7.17%, Dilijan 6.93%. Small regional towns occupy the top: Jermuk 8.68%, Gavar 8.27%, Stepanavan 8.04%, but with wider intervals reflecting thinner markets. The national baseline, $12 \exp(\mu_{\alpha,R}) / \exp(\mu_{\alpha,P})$, is $\sim 6.78\%$.

5.7 Naive median vs. model-based yield

Comparing the model-based yields with the raw $12 \cdot \text{median}(R) / \text{median}(P)$ calculation per district reveals where composition adjustment matters most. In Gyumri the raw ratio is 5.63% vs. a model estimate of 7.17%; in Proshyan it is 3.80% vs. 6.12%; in Jermuk, conversely, 10.44% vs. 8.68% (Figure 6). The direction of correction depends on whether a district’s rental pool is unusually differentiated relative to its sale pool, or vice versa. The model produces a much narrower, smoother spread of estimated yields than naive medians do, showing that the standardization is removing real composition noise rather than imposing artificial smoothness.

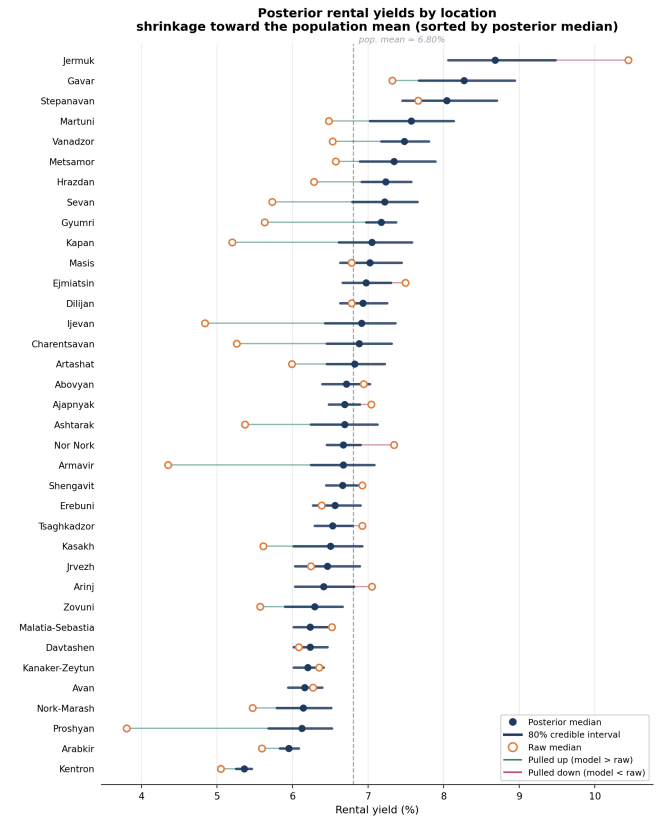


Figure 6: Raw median yield vs. model-based standardized yield, by district.

6 Discussion and Conclusion

The Armenia demonstration produces a set of internally consistent and substantively plausible findings. Implied gross yields lie within a tight 5.4–8.7% band, narrower and more orderly than the noisy 3.8–10.4% raw median range. Smaller apartments offer higher gross yields, consistent with sale prices rewarding size more aggressively than rents do. Observed condition differences are capitalized more strongly in the rental market than in the resale market: the lowest-condition units, despite being cheap to buy, are estimated to be the lowest-yielding because their rent discount exceeds their price discount.

Methodologically, the framework demonstrates that a country-scale yield surface can be built from public online listings without matched-pair data, an official transaction register, or proprietary survey panels. Where rent and sale observations refer to different dwellings, the multivariate hierarchical structure recovers location-level yields by leveraging the strong ($\rho \approx 0.99$) connection between rent and sale premia. The modeling structure is transferable to other markets with sufficiently deep classifieds data, although each platform would require market-specific scraping, duplicate handling, currency treatment, and feature harmonization.

For economic monitoring, the main value of the framework is measurement: it converts large, imperfect online listing

pools into a composition-adjusted rent–price signal with explicit uncertainty. Such a measure could complement conventional house-price indices, rental statistics, and credit-market indicators in settings where matched housing microdata are scarce. The posterior yield surface can also be combined with additional assumptions about vacancy, financing, taxes, maintenance, and transaction costs to study related quantities such as net yield gaps or user-cost comparisons across primary and secondary cities. For lenders and market participants, the same framework identifies which sub-markets price rental income generously relative to capital outlay. For the housing-economics literature, it provides a sub-national yield estimand that remains operational when rent–sale records cannot be physically matched.

The high ρ deserves a final substantive comment. A near-perfect rent–sale correlation across locations means that, to first order, “where rent is high, price is high.” The interesting variation lies in the second-order deviations from that collinearity, where a district’s sale price is higher than the rent surface would predict, and vice versa. The MVN parameterization isolates exactly those deviations as the source of cross-location yield differences.

7 Limitations

Listing-data limitations. The estimates are based on online listing data rather than completed transactions. Advertised rents and sale prices may include seller price anchoring, negotiation buffers, and platform-specific listing behavior. The snapshot may also contain duplicate, stale, or strategically reposted listings if agents relist the same dwelling. The data also come from a single classifieds platform over a short cross-sectional window, so they cannot identify seasonality, time trends, or supply shocks. Accordingly, the present application should be interpreted as a cross-sectional measurement exercise rather than a high-frequency monitoring or forecasting product. Finally, renovation and furniture fields are self-reported listing attributes. Measurement error in these fields may impact the corresponding coefficients and increase residual variation, although the main yield estimates remain anchored in observed rent and sale listings on both sides of each retained location.

Gross advertised yield, not net realized return. The estimand is a gross advertised yield: annualized asking rent divided by asking sale price for a standardized dwelling profile. It does not account for vacancy, property taxes, insurance, maintenance, management fees, transaction costs, financing, or negotiated discounts. The resulting estimates are therefore best interpreted as comparable market-level yield signals, not investor-level net returns. Estimating net yields would require additional location-specific cost layers beyond the listing data used here.

Model scope and extensions. The main specification deliberately uses a small set of widely reported dwelling characteristics: location, size, room count, renovation, and furniture. Additional fields such as floor, building type, balcony, coordi-

nates, and free-text descriptions could improve fit, but each would require careful harmonization across rent and sale listings to avoid reintroducing composition bias. The current model also does not exploit spatial dependence across nearby districts; a spatial extension using coordinates or neighborhood adjacency could tighten estimates in sparse locations. Finally, the posterior for ρ concentrates near one, which makes the bivariate covariance geometry harder to sample and lowers effective sample sizes for some top-level hyperparameters.

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Table 4: Posterior means and 94% intervals.

Parameter	Mean	3%	97%
$\mu_{\alpha,R}$	6.191	6.095	6.302
$\mu_{\alpha,P}$	11.368	11.251	11.506
$\sigma_{\alpha,R}$	0.324	0.252	0.399
$\sigma_{\alpha,P}$	0.409	0.320	0.502
ρ	0.984	0.966	0.998
β_{sqm}^R	0.521	0.485	0.560
β_{rm}^R	0.019	0.002	0.036
σ_R	0.255	0.250	0.261
β_{sqm}^P	0.904	0.875	0.935
β_{rm}^P	-0.035	-0.048	-0.023
σ_P	0.250	0.245	0.254
<i>Renovation (vs. Major)</i>			
$\beta_{\text{Designer}}^R$	0.130	0.112	0.150
β_{Euro}^R	-0.082	-0.103	-0.060
$\beta_{\text{Cosmetic}}^R$	-0.310	-0.342	-0.275
β_{None}^R	-0.542	-0.716	-0.375
β_{Old}^R	-0.509	-0.578	-0.440
β_{Partial}^R	-0.394	-0.461	-0.331
$\beta_{\text{Designer}}^P$	0.104	0.079	0.128
β_{Euro}^P	-0.065	-0.087	-0.043
$\beta_{\text{Cosmetic}}^P$	-0.189	-0.213	-0.167
β_{None}^P	-0.133	-0.159	-0.107
β_{Old}^P	-0.283	-0.308	-0.255
β_{Partial}^P	-0.185	-0.213	-0.158
<i>Furniture (vs. With)</i>			
β_{Without}^R	-0.128	-0.219	-0.036
β_{Partial}^R	-0.131	-0.184	-0.084
$\beta_{\text{By Agreement}}^R$	-0.166	-0.284	-0.048
β_{Without}^P	-0.053	-0.075	-0.031
β_{Partial}^P	-0.073	-0.093	-0.053
$\beta_{\text{By Agreement}}^P$	-0.070	-0.093	-0.048

A Full coefficient table

Table 4 reports posterior means and 94% credible intervals for all regression coefficients and the variance components. Renovation effects are read against the reference Major; furniture effects against With.

B Full posterior summary and sampler diagnostics

Table 5 reports posterior summaries, Monte Carlo standard errors, effective sample sizes, and $\hat{R}$ diagnostics for the parameters.

C Per-location yield posterior

Table 6 reports posterior median yields and 80% credible intervals at the baseline profile (60 m², 2-room, Major, With) for all 36 retained locations, sorted descending.

Table 5: Full posterior summary and sampler diagnostics.

Parameter	Mean	SD	HDI 3%	HDI 97%	MCSE mean	MCSE SD	ESS bulk	ESS tail	$\hat{R}$
$\mu_{\alpha, R}$	6.191	0.055	6.095	6.302	0.003	0.002	254.487	674.215	1.013
$\mu_{\alpha, P}$	11.368	0.068	11.251	11.506	0.004	0.002	254.136	672.100	1.013
$\text{corr}_{\alpha}[R, R]$	1.000	0.000	1.000	1.000	0.000	—	4000.000	4000.000	—
$\text{corr}_{\alpha}[R, P]$	0.984	0.010	0.966	0.998	0.000	0.000	1018.915	1903.604	1.002
$\text{corr}_{\alpha}[P, R]$	0.984	0.010	0.966	0.998	0.000	0.000	1018.915	1903.604	1.002
$\text{corr}_{\alpha}[P, P]$	1.000	0.000	1.000	1.000	0.000	0.000	4040.165	3278.660	1.000
$\sigma_{\alpha, R}$	0.324	0.040	0.252	0.399	0.002	0.001	575.544	861.068	1.006
$\sigma_{\alpha, P}$	0.409	0.050	0.320	0.502	0.002	0.001	572.398	829.744	1.006
ρ	0.984	0.010	0.966	0.998	0.000	0.000	1018.915	1903.604	1.002
β_{sqm}^R	0.521	0.020	0.485	0.560	0.000	0.000	3017.387	2699.035	1.000
β_{rooms}^R	0.019	0.009	0.002	0.036	0.000	0.000	2866.017	2696.092	1.001
σ_R	0.255	0.003	0.250	0.261	0.000	0.000	7470.129	2737.876	1.000
β_{sqm}^P	0.904	0.016	0.875	0.935	0.000	0.000	3497.153	3015.346	1.001
β_{rooms}^P	-0.035	0.007	-0.048	-0.023	0.000	0.000	3540.809	3182.356	1.001
σ_P	0.250	0.002	0.245	0.254	0.000	0.000	5833.431	2756.426	1.004
$\beta_{\text{Designer}}^R$	0.130	0.010	0.112	0.150	0.000	0.000	4137.126	3423.925	0.999
β_{Euro}^R	-0.082	0.011	-0.103	-0.060	0.000	0.000	3935.893	2833.203	1.000
$\beta_{\text{Cosmetic}}^R$	-0.310	0.018	-0.342	-0.275	0.000	0.000	6607.983	3333.308	1.001
β_{None}^R	-0.542	0.091	-0.716	-0.375	0.001	0.001	4815.271	2898.750	1.000
β_{Old}^R	-0.509	0.037	-0.578	-0.440	0.000	0.001	5652.030	3248.508	1.002
β_{Partial}^R	-0.394	0.035	-0.461	-0.331	0.000	0.001	4874.664	2940.799	1.000
$\beta_{\text{Without furn.}}^R$	-0.128	0.048	-0.219	-0.036	0.001	0.001	5564.365	2805.689	1.001
$\beta_{\text{Partial furn.}}^R$	-0.131	0.027	-0.184	-0.084	0.000	0.000	5034.365	3115.699	1.000
$\beta_{\text{By agreement}}^R$	-0.166	0.063	-0.284	-0.048	0.001	0.001	6287.324	2910.220	1.000
$\beta_{\text{Designer}}^P$	0.104	0.013	0.079	0.128	0.000	0.000	4789.952	3090.514	1.002
β_{Euro}^P	-0.065	0.012	-0.087	-0.043	0.000	0.000	4100.702	3193.827	1.000
$\beta_{\text{Cosmetic}}^P$	-0.189	0.012	-0.213	-0.167	0.000	0.000	3920.064	2850.800	1.002
β_{None}^P	-0.133	0.014	-0.159	-0.107	0.000	0.000	2908.315	2721.347	1.001
β_{Old}^P	-0.283	0.014	-0.308	-0.255	0.000	0.000	3913.019	3023.083	1.002
β_{Partial}^P	-0.185	0.015	-0.213	-0.158	0.000	0.000	3856.828	3165.101	1.001
$\beta_{\text{Without furn.}}^P$	-0.053	0.012	-0.075	-0.031	0.000	0.000	3162.730	3064.366	1.000
$\beta_{\text{Partial furn.}}^P$	-0.073	0.011	-0.093	-0.053	0.000	0.000	4082.957	2458.183	1.001
$\beta_{\text{By agreement}}^P$	-0.070	0.012	-0.093	-0.048	0.000	0.000	6264.897	3167.198	1.000

Table 6: Implied gross yield (%) at baseline profile, all retained locations.

Location	Median	10%	90%
Jermuk	8.68	8.06	9.48
Gavar	8.27	7.67	8.94
Stepanavan	8.04	7.45	8.70
Martuni	7.57	7.02	8.13
Vanadzor	7.48	7.17	7.80
Metsamor	7.34	6.89	7.89
Hrazdan	7.23	6.91	7.57
Sevan	7.22	6.79	7.65
Gyumri	7.17	6.97	7.37
Kapan	7.05	6.61	7.58
Masis	7.02	6.63	7.44
Ejmiatsin	6.97	6.66	7.30
Dilijan	6.93	6.63	7.25
Ijevan	6.91	6.43	7.36
Charentsavan	6.88	6.45	7.31
Artashat	6.82	6.45	7.22
Abovyan	6.71	6.39	7.02
Ashtarak	6.69	6.24	7.12
Ajapnyak	6.69	6.48	6.89
Armavir	6.67	6.24	7.08
Nor Nork	6.67	6.45	6.90
Shengavit	6.66	6.44	6.90
Erebuni	6.56	6.27	6.90
Tsaghkadzor	6.53	6.29	6.79
Kasakh	6.50	6.01	6.92
Jrvezh	6.46	6.03	6.89
Arinj	6.41	6.03	6.81
Zovuni	6.29	5.90	6.66
Davtashen	6.23	6.01	6.46
Malatia-Sebastia	6.23	6.01	6.46
Kanaker-Zeytun	6.20	6.01	6.41
Avan	6.16	5.94	6.39
Nork-Marash	6.14	5.79	6.51
Proshyan	6.12	5.68	6.52
Arabkir	5.95	5.83	6.08
Kentron	5.36	5.25	5.46